Practice Set B

Class 12 Mathematics, Chapter 6: Application of Derivatives

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, Board-pattern — 1 mark

The radius of a right circular cylinder is increasing at the rate of 3 cm/s and its height is decreasing at the rate of 4 cm/s. The rate of change of its volume when the radius is 4 cm and the height is 6 cm is:

(A) $80\pi \text{ cm}^3/\text{s}$ **(B)** $144\pi \text{ cm}^3/\text{s}$ **(C)** $208\pi \text{ cm}^3/\text{s}$ **(D)** $64\pi \text{ cm}^3/\text{s}$

2 MCQ, NCERT Exemplar — 1 mark

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + \cos x$. Then f:

(A) has a minimum at $x = \pi$ **(B)** has a maximum at $x = 0$

(C) is a decreasing function **(D)** is an increasing function

3 MCQ, CBSE SQP 2026-27 — 1 mark

If $y = a\left\{x + 2a\sin\left(\frac{x}{a}\right)\right\}$, $a > 0$, $x \in [0, \pi a]$, is a decreasing function, then the interval in which x lies is:

(A) $\left(\frac{5\pi}{6}, \pi\right]$ **(B)** $\left(\frac{2\pi}{3}, \pi\right]$ **(C)** $\left(\frac{5\pi a}{6}, \pi a\right]$ **(D)** $\left(\frac{2\pi a}{3}, \pi a\right]$

4 MCQ, NCERT Exemplar — 1 mark

At $x = \frac{5\pi}{6}$, $f(x) = 2\sin 3x + 3\cos 3x$ is:

(A) maximum **(B)** minimum

(C) zero **(D)** neither maximum nor minimum

5 MCQ, Board-pattern — 1 mark

The local minimum value of $f(x) = x^x$, $x > 0$, is:

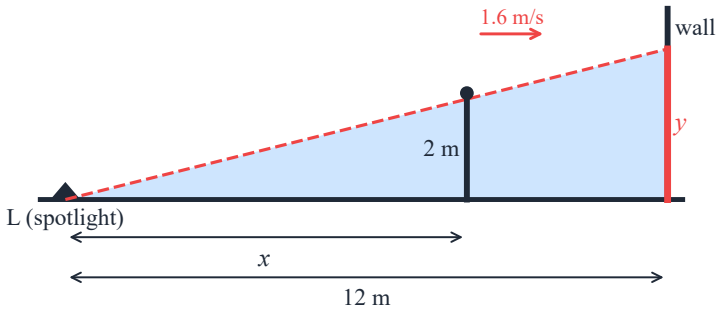
(A) $e^{-1/e}$ **(B)** $e^{1/e}$ **(C)** $\frac{1}{e}$ **(D)** 1

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** $f(x) = x^3$ is strictly increasing on $\mathbb{R}$. **Reason (R):** If a function f is strictly increasing on an interval, then $f'(x) > 0$ for every x in that interval. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, NCERT Ex 6.1 Q2 The volume of a cube is increasing at the rate of $8\ \text{cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 12 cm?	2 marks

Level 2

19 marks

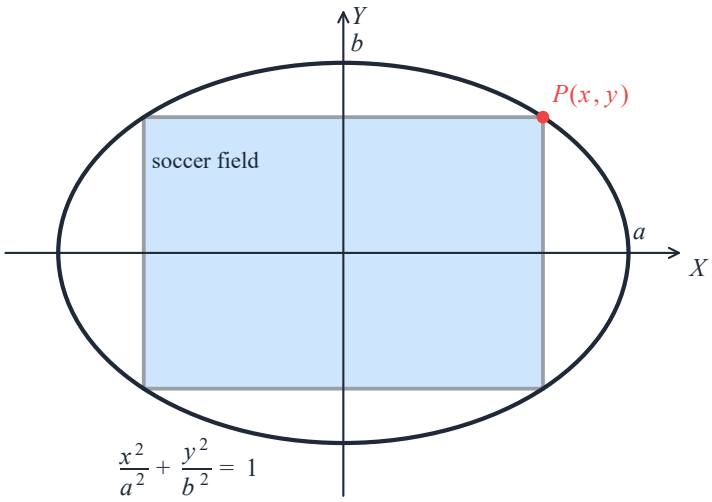
8	Short answer, PYQ-style Find the intervals in which the function $f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12$ is (a) strictly increasing (b) strictly decreasing.	3 marks
9	Short answer, PYQ-style A spherical balloon is being inflated so that its volume increases at the rate of $25\ \text{cm}^3/\text{s}$. Find the rate at which its surface area is increasing when the radius is 5 cm.	2 marks
10	Short answer, NCERT Exemplar Find the points of local maxima, local minima and the points of inflexion (critical points where f' does not change sign) of the function $f(x) = x^5 - 5x^4 + 5x^3 - 1$. Also find the corresponding local maximum and local minimum values.	3 marks
11	Short answer, PYQ-style Find the intervals in which the function $f(x) = \log(1 + x) - \frac{x}{1 + x}$, $x > -1$, is strictly increasing or strictly decreasing.	3 marks
12	Short answer, CBSE SQP 2026-27 The circular area of vision on the ground by a surveillance drone fitted with a camera increases at the rate of $44\ \text{m}^2$ per metre as it gains vertical height. The radius of the circle on the ground increases at the rate of 6 m/s. Find the rate at which the height of the drone increases with time, at the instant when the radius of the circular area on the ground is 14 m. (Use $\pi = \frac{22}{7}$)	3 marks
13	Short answer, NCERT Misc. Ex. Q11 Find the absolute maximum and minimum values of the function f given by $f(x) = \cos^2 x + \sin x$, $x \in [0, \pi]$.	3 marks
14	Short answer, NCERT Ex 6.3 Q14 Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.	2 marks

Q No.	Question	Marks
Level 3		**25 marks**
15	Long answer, PYQ-style Show that the isosceles triangle of maximum area that can be inscribed in a circle of radius r is an equilateral triangle. Also find its maximum area.	5 marks
16	Long answer, NCERT Misc. Ex. Q15 Show that the height of the cylinder of greatest volume which can be inscribed in a right circular cone of height h and semi-vertical angle α is one-third that of the cone and the greatest volume of the cylinder is $\frac{4}{27}\pi h^3 \tan^2 \alpha$.	5 marks
17	Long answer, NCERT Ex 6.3 Q22 A wire of length 28 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a circle. What should be the length of the two pieces so that the combined area of the square and the circle is minimum?	5 marks
18	Short answer, NCERT Misc. Ex. Q3 Find the intervals in which the function f given by $f(x) = \frac{4\sin x - 2x - x\cos x}{2 + \cos x}$, $0 \le x \le 2\pi$, is (i) increasing (ii) decreasing.	3 marks
19	Short answer, Board-pattern A spotlight on the ground shines on a vertical wall 12 m away. A man 2 m tall walks from the spotlight straight towards the wall at a speed of 1.6 m/s. How fast is the height of his shadow on the wall decreasing when he is 4 m from the wall? 	3 marks

Q No.	Question	Marks

20 Case study, CBSE SQP 2022-23 — 4 marks

Case Study: In an elliptical sports field, the authority wants to design a rectangular soccer field with the maximum possible area. The sports field is given by the graph of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.



(i) If the length and the breadth of the rectangular field are $2x$ and $2y$ respectively, find the area function in terms of x. — 1 mark

(ii) Find the critical point of the function. — 1 mark

(iii)(a) Use the first derivative test to find the length $2x$ and width $2y$ of the soccer field (in terms of a and b) that maximise its area. — 2 marks

OR

(iii)(b) Use the second derivative test to find the length $2x$ and width $2y$ of the soccer field (in terms of a and b) that maximise its area. — 2 marks

Answer key

Chapter 6: Application of Derivatives, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $80\pi \text{ cm}^3/\text{s}$
2	(D) is an increasing function
3	(D) $\left(\frac{2\pi a}{3}, \pi a\right]$
4	(D) neither maximum nor minimum
5	(A) $e^{-1/e}$
6	(C) A is true, but R is false.
7	Surface area increases at $\frac{8}{3} \text{ cm}^2/\text{s}$

Level 2

8	Strictly increasing on $(-3, 2) \cup (4, \infty)$; strictly decreasing on $(-\infty, -3) \cup (2, 4)$
9	Surface area increases at $10 \text{ cm}^2/\text{s}$
10	Local maximum at $x = 1$, value 0; local minimum at $x = 3$, value -28; point of inflexion at $x = 0$
11	Strictly decreasing on $(-1, 0)$; strictly increasing on $(0, \infty)$
12	The height of the drone increases at 12 m/s
13	Absolute maximum $= \frac{5}{4}$ (at $x = \frac{\pi}{6}, \frac{5\pi}{6}$); absolute minimum $= 1$ (at $x = 0, \frac{\pi}{2}, \pi$)
14	$x = 15, y = 45$

Level 3

15	Maximum area when all sides are $\sqrt{3}\,r$ (equilateral); maximum area $= \frac{3\sqrt{3}}{4} r^2$
16	Height of cylinder $= \frac{h}{3}$; greatest volume $= \frac{4}{27}\pi h^3 \tan^2\alpha$
17	Square piece $= \frac{112}{\pi + 4}$ m and circle piece $= \frac{28\pi}{\pi + 4}$ m
18	(i) Increasing on $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$ (ii) decreasing on $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$
19	The shadow on the wall is decreasing at 0.6 m/s
20	(i) $A(x) = \frac{4b}{a} x\sqrt{a^2 - x^2}$ (ii) $x = \frac{a}{\sqrt{2}}$ (iii) length $\sqrt{2}\,a$, width $\sqrt{2}\,b$ (maximum area $2ab$)